

From Root Cause to AI Agents: Elastic & Gigamon in Action

GIGAMON SESSION

Finding the Bottleneck: Root-Cause Analysis with Elastic and Gigamon

DESCRIPTION

Modern applications fail across complex service chains, but **observability** data alone may not reveal where latency begins. In this technical session, **Gigamon** and Elastic show how **network-derived intelligence** and **application metadata** can enrich **Elastic Observability**. Learn to map client-to-app dependencies, separate network, application, HTTP, and DNS latency, and isolate degraded service edges across hybrid cloud and Kubernetes. Leave with a practical framework for faster, evidence-based root-cause analysis and clearer ownership across network, platform, SRE, and application teams.

Hiding in Network Traffic

Network-Derived Evidence to detect, troubleshoot, and secure.

Roman Roytfeld | Sr Sales Engineer @ Gigamon





In 2004, Gigamon created the packet-broker category

As networks and monitoring-tool demands grew complex, the mission was simple: make the right traffic available to the right tools.



The original mission: getting the right network traffic to the right tools.

In 2019, Gigamon added application intelligence to the access layer

Classic network log (5 Tuple)

- NetFlow
- Gigamon deep observability (Network derived Telemetry)

5000+ Attributes

- Mac Address, DHCP options
- Ingress interface (if available via SNMP)

Source IP,

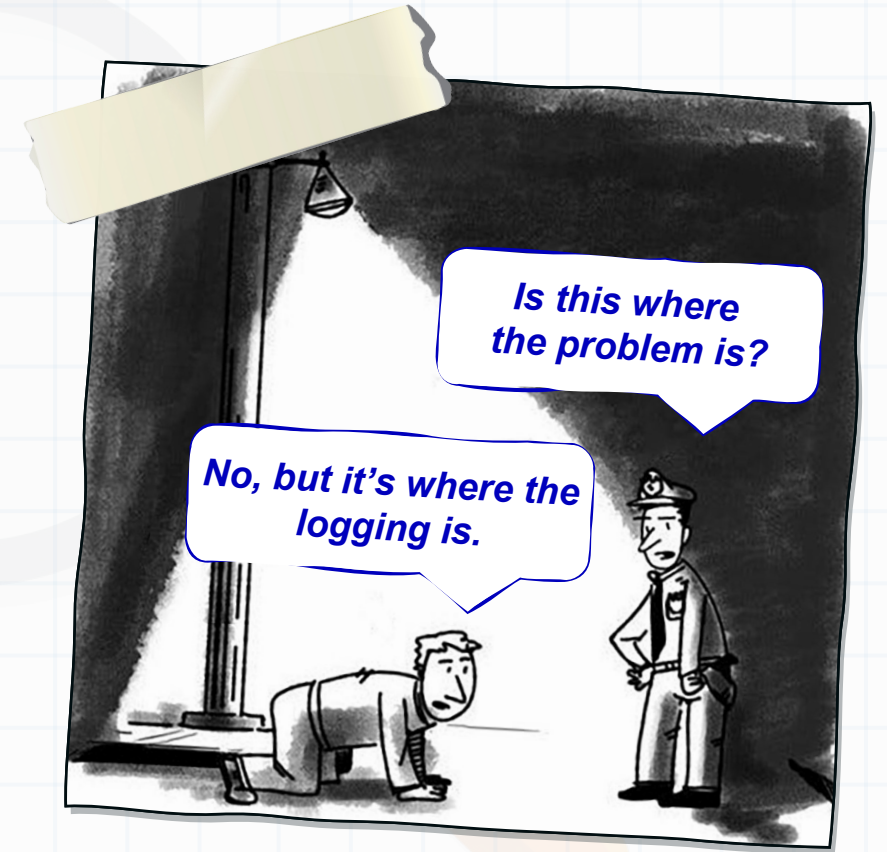
Source port,

Destination IP,

Destination port,

Specific protocol (TCP/UDP)

- Flow start time
- Protocols information (buffer size, retransmits, errors)
- Encryption levels
- Certificate information
- Layer 6 encoding
- Application visibility and detection



Deployed by Fortune 100 companies and government agencies worldwide

4,000+

Joint Elastic + Gigamon
customers worldwide

15+

Pre-built dashboards
out of the box

Day 0

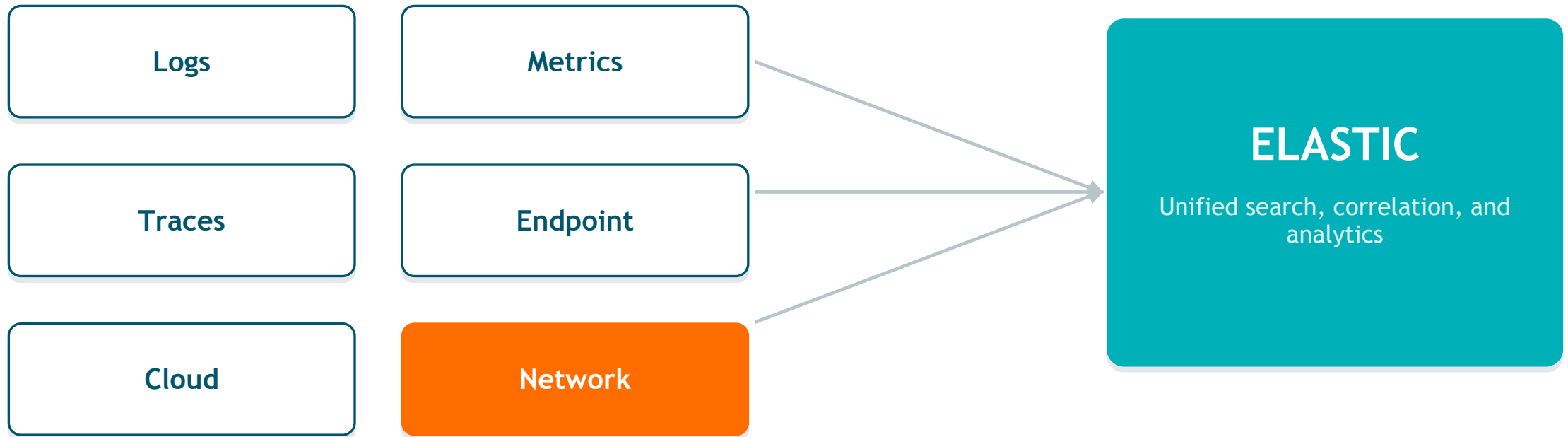
Value with the
certified integration

"It's the difference between looking through a keyhole and an open door. With Gigamon and Elastic, we finally see the full picture."

— Gareth Maclachlan, COO, Gigamon

Elastic is the natural place to unify network evidence

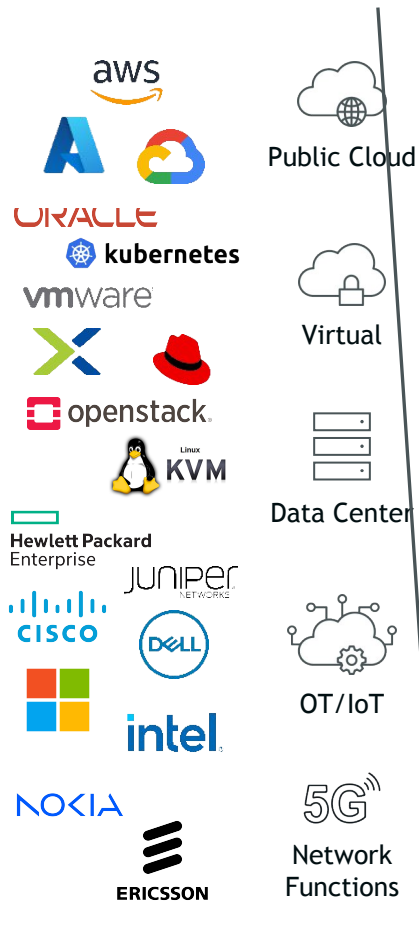
Security, NetOps, DevOps, and SRE teams already correlate diverse operational data in Elastic.



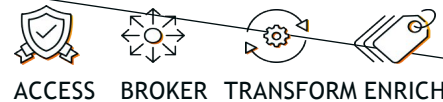
A shared data layer lets every team investigate the same event from multiple perspectives — network data adds the “how” behind endpoint, application, and cloud records.

Network-Derived Evidence to detect, troubleshoot, and secure

Workloads and Platforms



Network Data In



Enriched Metadata
Fueling Your AI

AI / Data Tools

SIEM
Security Info. & Event Mgmt

Observability
Observability Tools

Agentic AI
Platforms & AI Agents

Data Lake
Storage & Analytics

Data Engines
And other Telemetry Pipelines



Packet Tools

NDR / XDR
Network Detection & Response

IDS / NSM
Intrusion Detection System

PCAP
Packet Capture

APM
App Perf. Mgmt

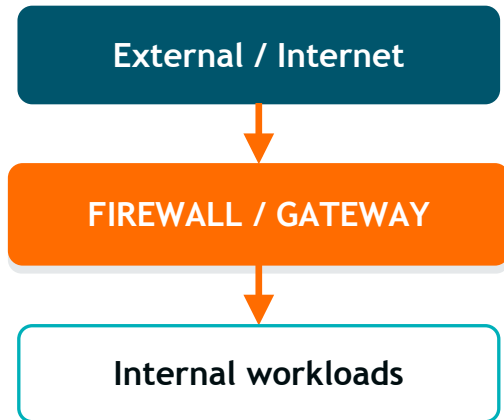
NPMD
Network Perf. Mgmt & Diagnostics



NetFlow lacks context and misses traffic

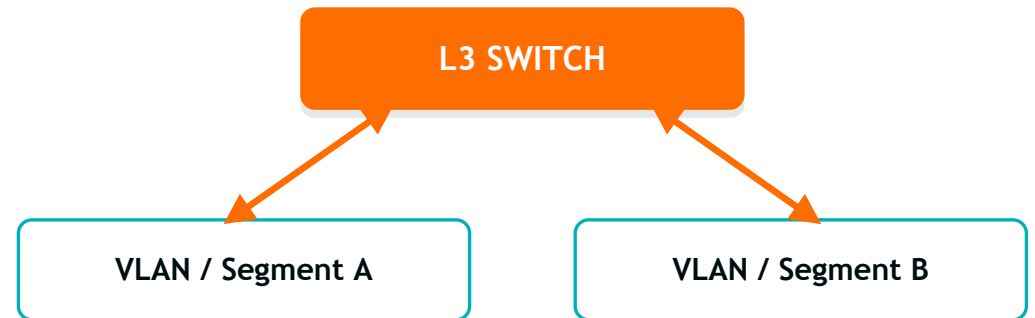
Firewall and internal switch are two observation points – some lateral and cloud paths still bypass both.

OBSERVATION POINT 1 · FIREWALL / GATEWAY



North-South traffic crossing the perimeter

OBSERVATION POINT 2 · INTERNAL L3 SWITCH



Routed East-West between VLANs, segments, or subnets

May bypass both observation points – **needs virtual-switch or cloud tapping**

Same-segment traffic

VM-to-VM

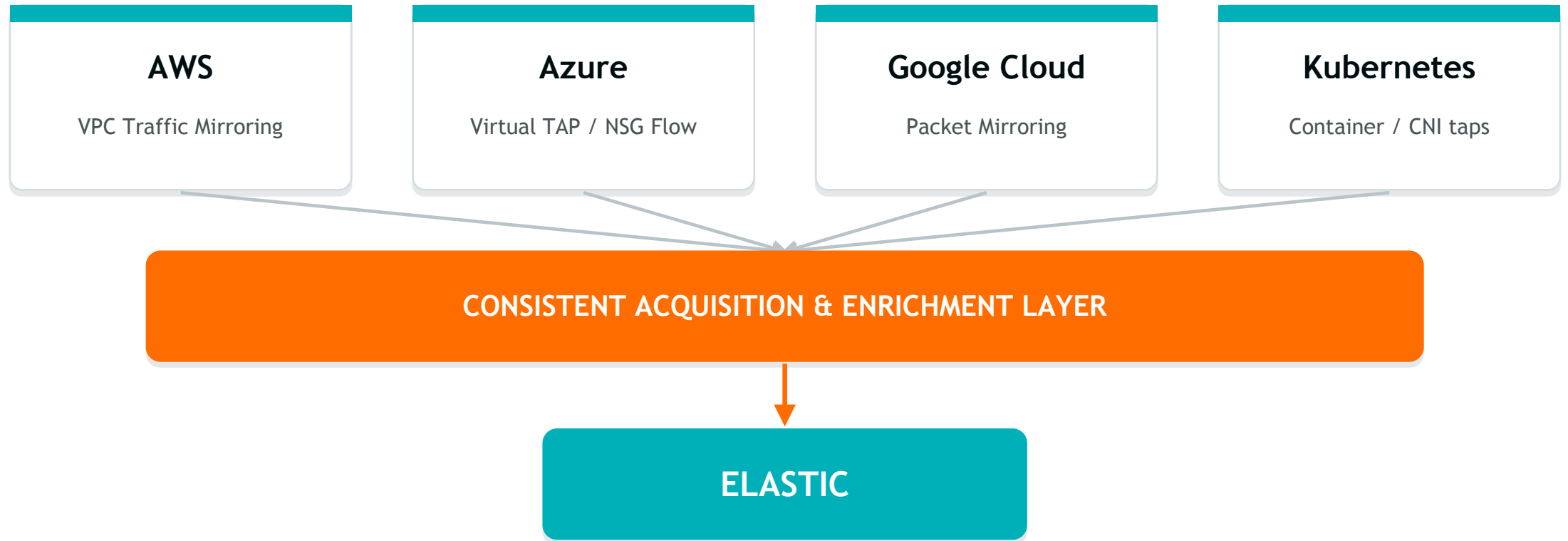
Container-to-container

Cloud paths

*At scale, switches may sample traffic or cap tracked flows – basic NetFlow stays primarily Layer 2-4 context and is sampled.
Netflow generation is best-effort.*

Cloud visibility requires a consistent access strategy

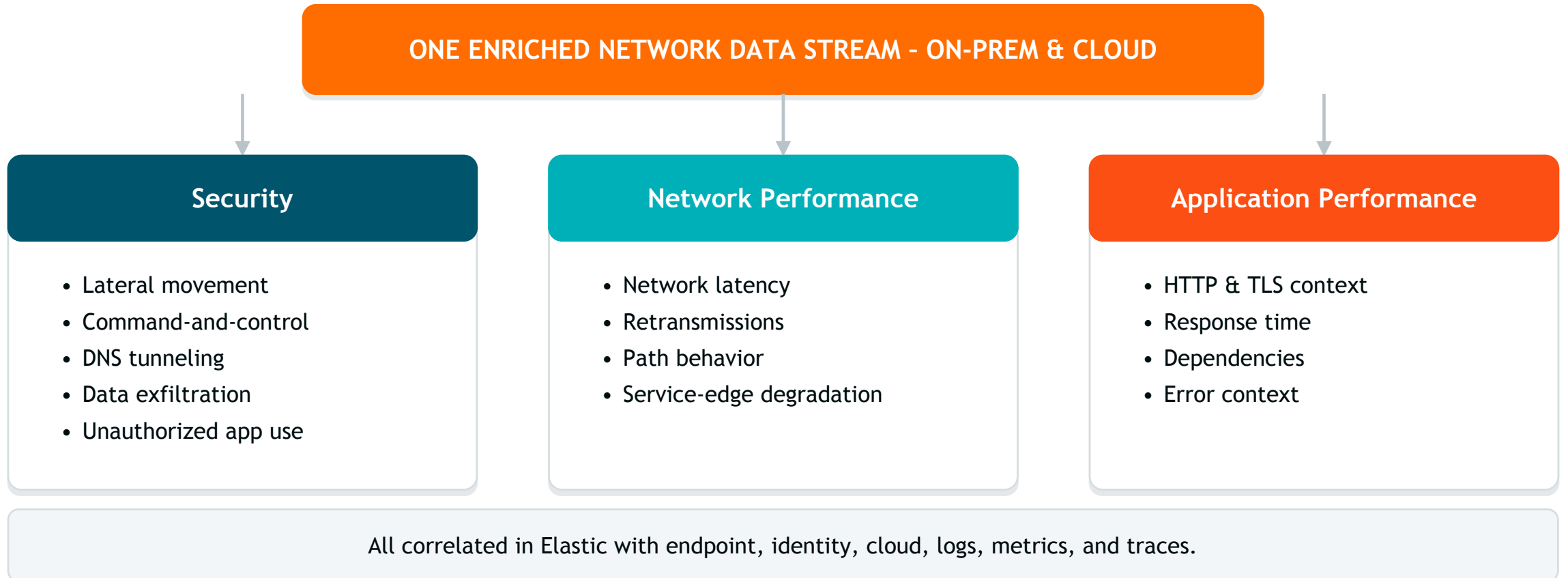
Every platform exposes traffic differently — a consistent layer keeps Elastic users covered.



East-West, intra-VM, container, and multi-tier traffic isn't visible by default.

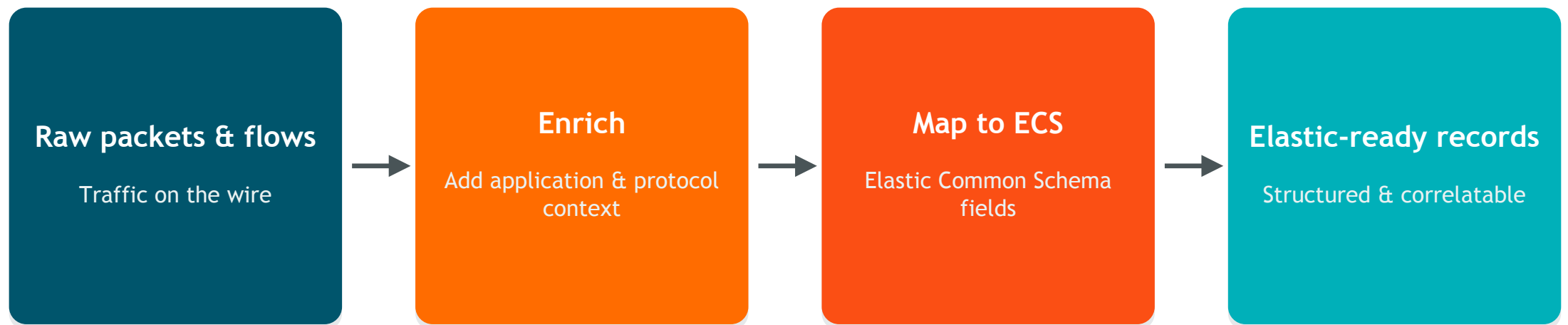
One enriched data stream supports three Elastic outcomes

The same application-aware evidence serves security, network, and application teams.



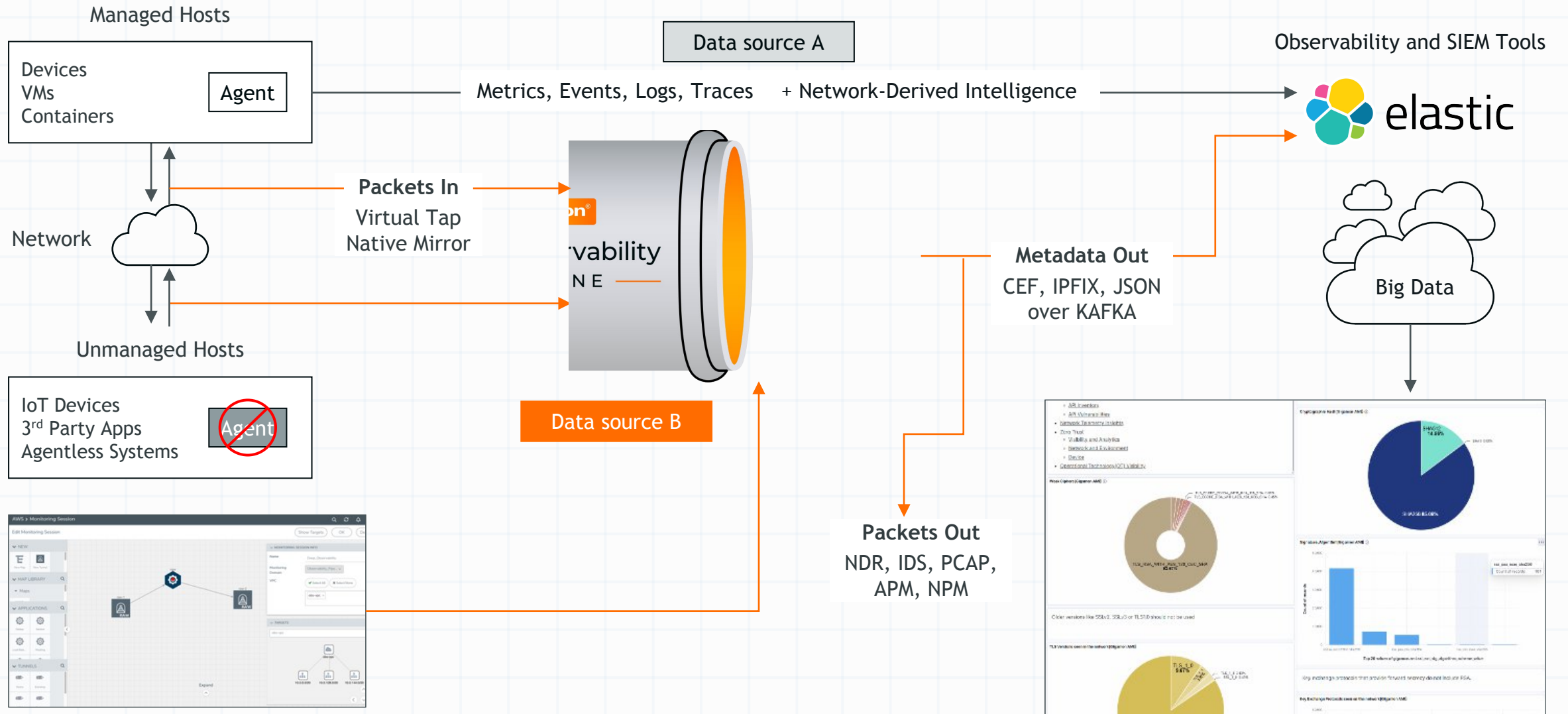
The right data format makes network intelligence useful

Searchable, correlatable, and compatible with the workflows Elastic users already run.



ECS mapping lets network-derived data work alongside every other Elastic data source – searchable, correlatable, and ready for existing dashboards and detections.

Share Deep Observability



Gigamon captures network truth. Elastic turns it into action.

GIGAMON

Deep Observability Pipeline

- ✓ 6,000+ app metadata attributes via deep packet inspection
- ✓ Layer 4-7 analysis without decryption
- ✓ 4,000+ applications identified in real time
- ✓ ECS-normalized output for instant ingestion
- ✓ Webhook delivery to Elastic Agent



ECS

ELASTIC

AI-Powered + Analytics

- ✓ 15+ pre-built dashboards — day-one value
- ✓ Correlate network + endpoint + cloud logs
- ✓ ML-powered anomaly detection at scale
- ✓ Investigations in hours, not days
- ✓ Attack detection, threat hunting, and response

Agentic AI built on network telemetry — natively integrated with Elastic Security

GIGAMON INSIGHTS

Agentic AI for network-derived telemetry

Gigamon Insights is an agentic AI application purpose-built for network-derived telemetry. Launched with native integrations into Elastic, it delivers instant AI-guided investigation without manually combing through dashboards.

"Accelerating investigations without manually combing through dashboard data."

— Gigamon, September 2025

Instant Threat Guidance

AI surfaces the most relevant network context for any alert — in seconds, not minutes.

Root-Cause Analysis

Correlate network flows to identify root cause across hybrid and multi-cloud environments.

Compliance Acceleration

Auto-generate evidence for PCI-DSS, HIPAA, and Zero Trust audits from network telemetry.

Key Use Cases

Ask any question about YOUR network

What you get:

- ✓ DNS queries & tunneling detection
- ✓ SMB file operations visibility
- ✓ SSL/TLS handshake analysis
- ✓ TCP performance metrics
- ✓ HTTP/S app-layer metadata
- ✓ SNMP & flow analysis



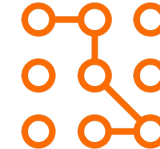
Security Insights

- ▶ Insecure protocols against policy
- ▶ Expired TLS certs / weak ciphers
- ▶ PQC compliance
- ▶ Lateral movement detection
- ▶ Authentication volume anomalies



Network Insights

- ▶ Latency between endpoints
- ▶ Traffic loss and network errors
- ▶ Source and destination volume trends
- ▶ DNS performance



Application Insights

- ▶ App response time and latency
- ▶ Network errors on specific apps
- ▶ Vulnerable API use
- ▶ Data center to cloud performance

Start Seeing More Today

Three ways to go deeper from this session:

01

Try the Integration

Live in your environment in under 30 minutes. Certified, supported, free to install.

02

Book a Live Demo

See Gigamon AML + Elastic in action on real network traffic. Ask us anything.

03

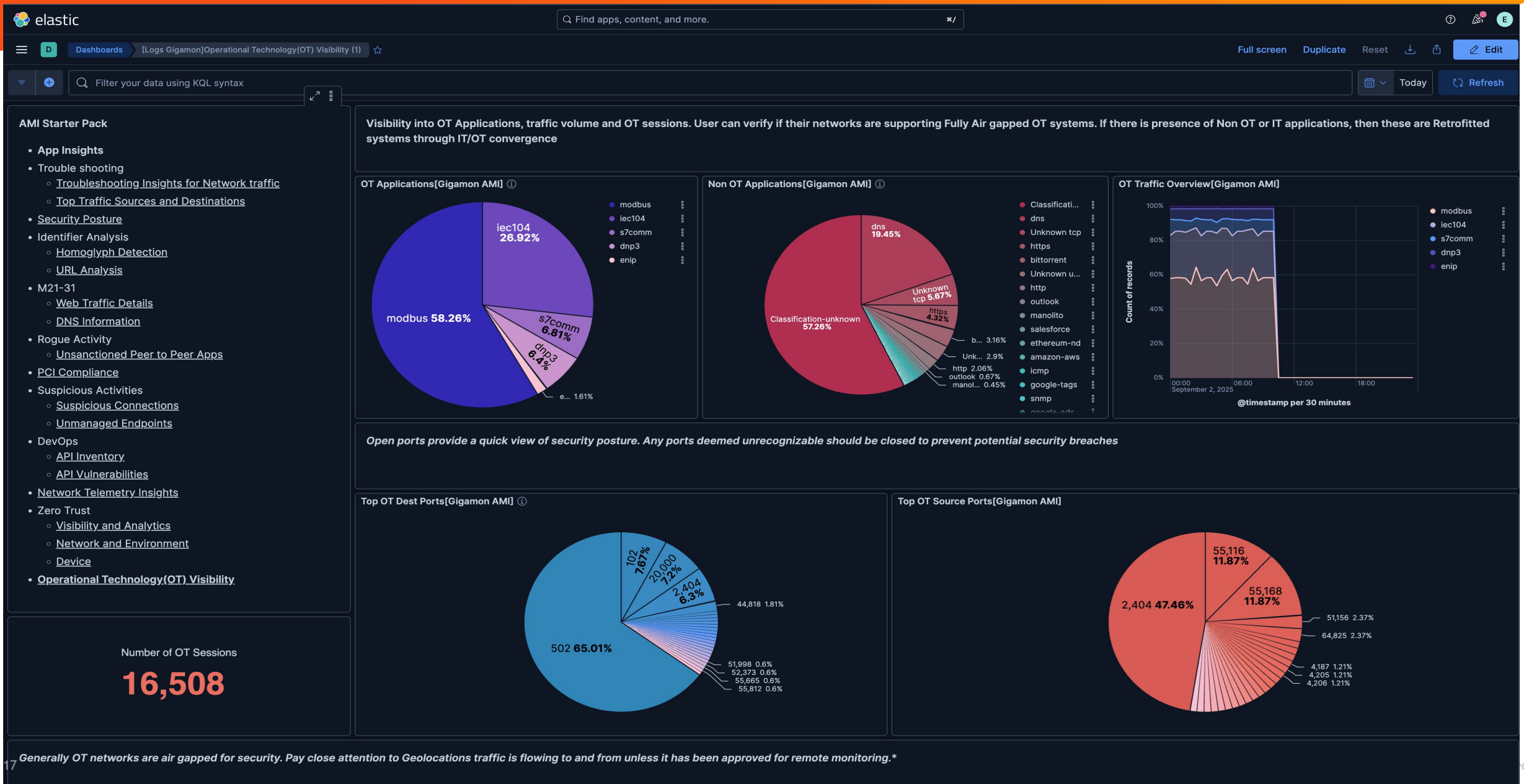
Let's chat

I am the guy with the orange shirt. You can't miss me.

elastic.co/docs/reference/integrations/gigamon



Elastic Partnership



Gigamon integrates with multiple tools. Here's what makes Elastic different.

ECS is the normalization target

Gigamon maps directly to Elastic Common Schema. Sentinel and SecOps require additional field mapping. Deploy in 30 minutes, not days.

Full-stack in one platform

Elastic Agent covers endpoint + cloud; Gigamon covers network. One licence, one console, one ES|QL query layer across all three sources.

Elastic Security Labs

In-house threat research team. Detection rules are written against ECS network events — the same schema Gigamon normalises to. Open-source, community-maintained.

Native ML anomaly detection

Built-in, not a separate licence. Unsupervised models score Gigamon network metadata alongside endpoint and cloud events in a single job.

ESL RESEARCH

Detection rules that run on ECS network events — the same schema Gigamon AML normalises to.

Hunting for Lateral Movement using EQL

SMB, RDP, RPC network event correlation — exactly the protocol metadata Gigamon AML surfaces.

elastic.co/security-labs/hunting-for-lateral-movement-using-event-query-language

Network Firewall Logs with Elastic Security

C2 comms, protocol anomalies, data exfiltration — detection patterns applicable to any ECS network source.

elastic.co/security-labs/make-the-most-of-network-firewall-logs-with-elastic

2025 Global Threat Report

1B+ events analyzed including network telemetry. Lateral movement is the #1 post-compromise technique.

elastic.co/resources/security/report/global-threat-report

Gigamon + Elastic Partnership

Integration details, ECS field mappings, and joint use case documentation.

elastic.co/blog/gigamon-partnership

Who is Gigamon

MARKET LEADERSHIP BACKED BY INDUSTRY RECOGNITION AND CUSTOMER VALIDATION



MARKET LEADERSHIP

#1

Deep Observability
Market Share Leader

51%

Market Share

650 Group, 2025



INDUSTRY RECOGNITION

FROST & SULLIVAN



2026 Frost & Sullivan
Company of the Year



Network Observability
for Cybersecurity

Frost & Sullivan, 2026



FEDERAL CIVILIAN CUSTOMER VALIDATION

Dept. of Homeland Security
Dept. of Energy
National Nuclear Security Admin
State Department
Internal Revenue Service
Health & Human Services
Social Security Administration
Dept. of Agriculture

FEDERAL CERTIFICATIONS

Gigamon Federal FCL
FIPS 140-3
IPV6
Common Criteria
CMMC Level 2 (in progress)

TRUSTED BY THE WORLD'S MOST DEMANDING ORGANIZATIONS
ACROSS SECURITY, OBSERVABILITY, IT OPERATIONS, AND AI

AI Traffic Intelligence

Gartner® Research: [CISOs Must Bring Shadow AI Into the Light for Secure Innovation](#)

79%

Of organizations suspect or have evidence that employees are misusing approved public GenAI



Monitor shadow AI use and engage with users to gain insights into objectives, data shared, benefits found and skills developed.”

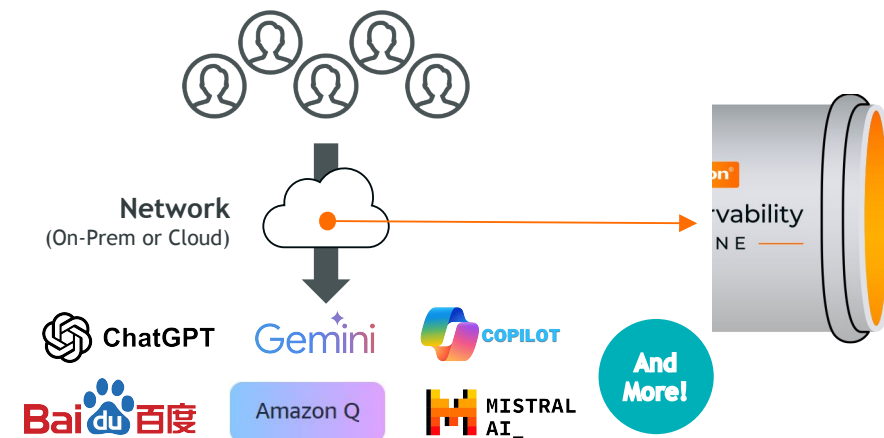
Source: Gartner, Andrew Walls, Jeremy D'Hoinne, John Watts, June 2025, GARTNER is a registered trademark and service mark of Gartner, Inc. and/or its affiliates in the U.S. and internationally and is used herein with permission. All rights reserved.

Gigamon®

Bring shadow AI usage into the light

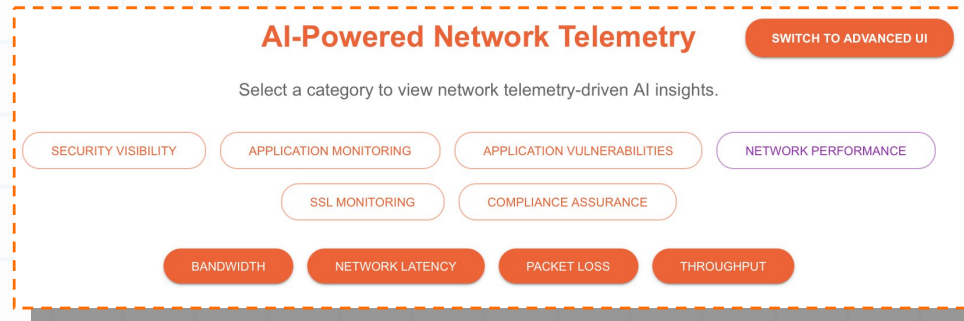
- ▶ Identify GenAI traffic, without disruption
- ▶ Spot GenAI embedded inside 3rd party applications
- ▶ Understand how GenAI tools are being used
- ▶ Discover GenAI usage across depts to optimize spend

Inform governance policy based on facts, not guesswork

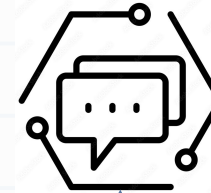


Network-Derived Metadata on GenAI

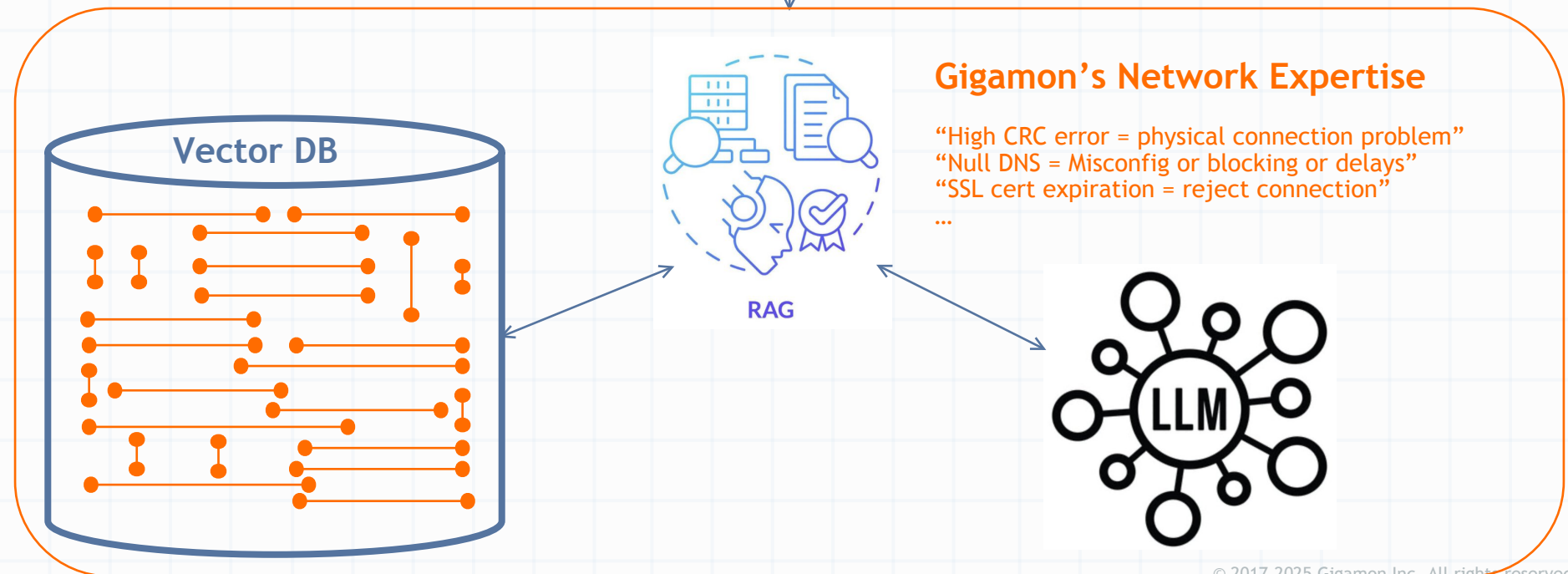
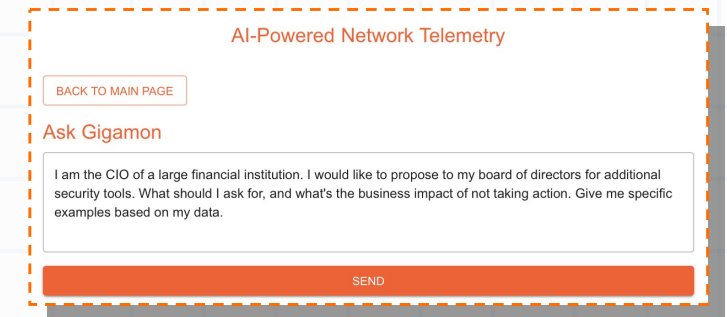
1. Pre-Programmed Prompts



MCP

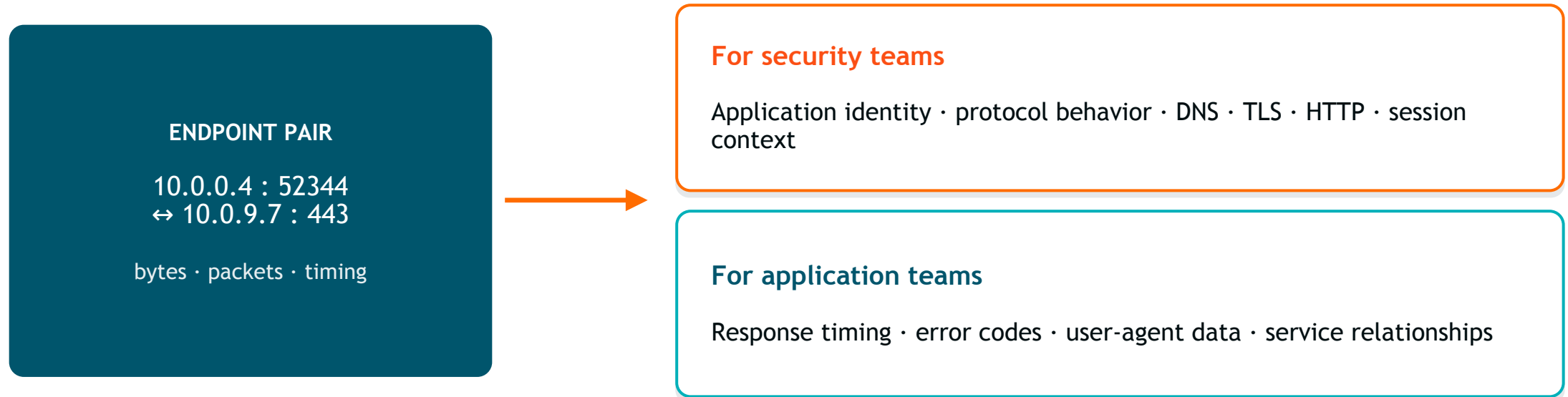


2. User Prompts



Network access should reveal more than endpoint pairs

Addresses, ports, and volume rarely explain whether behavior is legitimate, malicious, or inefficient.

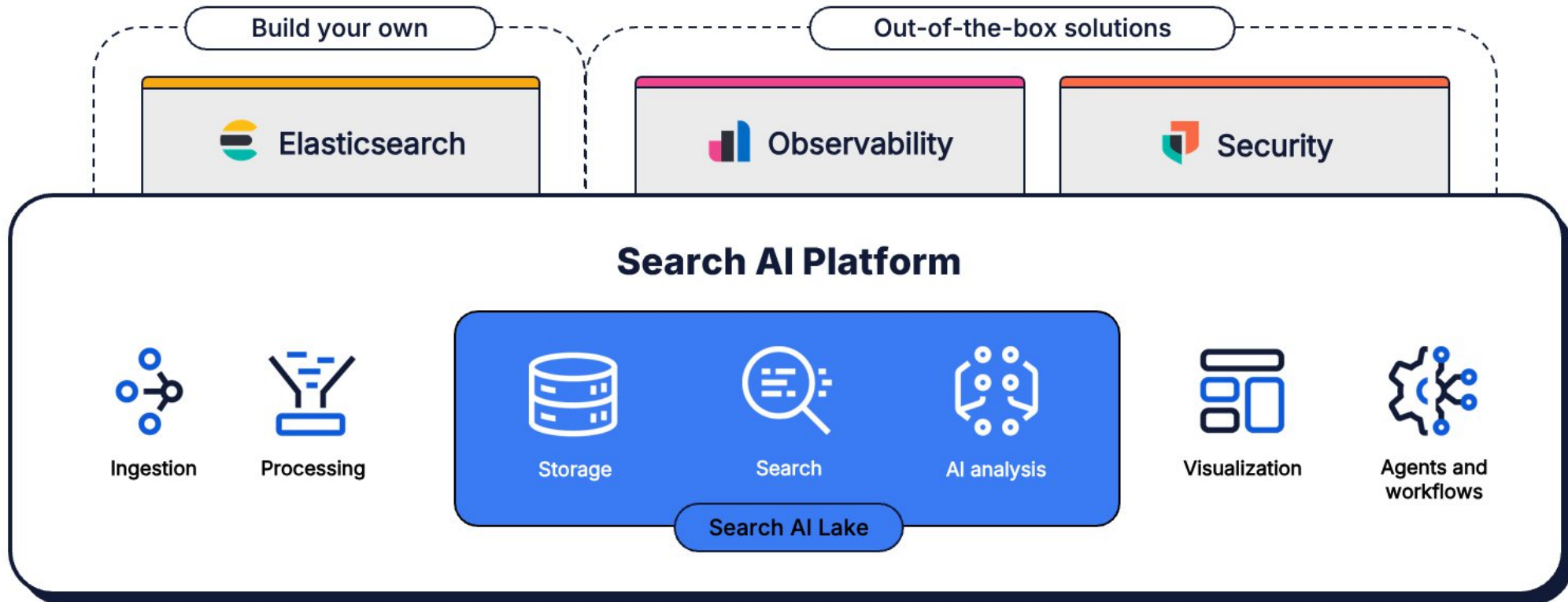


Network to detection in seconds



What is Elastic?

One platform, two out-of-the-box solutions, the freedom to build anything



The Gigamon Elastic integration



- Mapping of Gigamon AMI fields to the Elastic Common Schema (ECS)
 - ▶ Allows easier correlation of AMI streams to other sources ingested into Elastic
 - ▶ Allows Gigamon data to be immediately available to Elastic solutions (Security, Observability)
- Integration includes:
 - ▶ Pre-built custom visualizations and dashboards (“AMI Starter Packs”) in Elastic
 - ▶ Custom Gigamon mappings to pre-built Elastic integrations: DGA, Data Exfiltration Detection
- Ability to utilize Gigamon data streams in Elastic’s built-in Machine Learning application, AI Assistant, and within Agentic workflows

Q-DAY WILL BE BIGGER THAN Y2K —AND IT STARTS NOW

There is a non-Zero Chance Q day is already here

A cryptographically relevant quantum computer could break today's encryption **as soon as 2027**. Every byte of encrypted **data in motion** — and the **Department of War workflows** that depend on it — is already at risk.

Q-DAY: AS SOON AS 2027

The clock is not theoretical. **Serious focus is needed now** — migration to quantum-safe crypto takes years, not months.

THE ENIGMA PROBLEM

Like the **Enigma machine of WWII**, we may never know the moment encryption breaks. **Harvest now, decrypt later** is already happening.

THE IoT BLIND SPOT

Billions of **IoT / OT devices** ship with hard-coded, un-upgradeable crypto — a migration challenge with no easy patch.

SEE THE ENCRYPTED

You cannot protect what you cannot see. Visibility into **encrypted, lateral traffic** is the first move toward quantum readiness.

Identify and Verify PQC

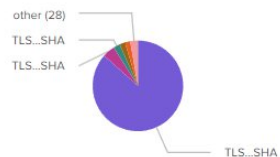
 **PQC Readiness: 0% (CRITICAL)**

Key Findings:

Category	Count	Percentage	Quantum Vulnerability Status
Quantum-Vulnerable Connections	63,028	100%	All connections use classical cryptography vulnerable to quantum attacks
PQC-Ready Connections	0	0%	No post-quantum cryptography detected
RSA Key Exchange (Quantum-Vulnerable)	61,346	97.3%	RSA will be broken by Shor's algorithm
ECDHE/DHE Key Exchange (Quantum-Vulnerable)	9,235	14.7%	Elliptic Curve & Diffie-Hellman vulnerable to quantum attacks
TLS 1.3 Capable	800	1.3%	TLS 1.3 required for PQC but no PQC algorithms in use
Legacy Protocol (SSL 3.0/TLS 1.0/1.1)	15,517	24.6%	Cannot support PQC at all

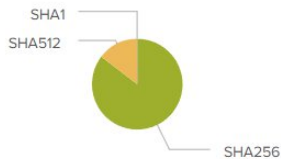
Weak Ciphers

Old or weak cryptographic algorithms or protocols used



Cryptographic Hash

Lookout for deprecated Hash functions such as MD5 and SHA1



TLS Versions seen in the network

Older versions like SSLv2, SSLv3 or TLS1.0 should not be used

